Questions

Q1.

Given that

$$(x-5)(2x+1)(x+3) \equiv ax^3 + bx^2 - 32x - 15$$

where a and b are constants,

(a) find the value of a and the value of b.

RocketRevise

(2)

(b) Hence find

$$\int \frac{(x-5)(2x+1)(x+3)}{5\sqrt{x}}\,dx$$

writing each term in simplest form.

(5)

(Total for question = 7 marks)

(**Q02 WMA11/01, Oct 2022**)

Q2.

A curve has equation $y = f(x)$.

www.RocketRevise.com

The point $P\left(4, \frac{32}{3}\right)$ lies on the curve.

Given that

- $f''(x) = \frac{4}{\sqrt{x}} - 3$

- $f'(x) = 5$ at P

compiled by Ahmed Gul www.RocketRevise.com

find

(a) the equation of the tangent to the curve at P, writing your answer in the form $y = mx + c$, where m and c are constants to be found,

(2)

(b) f(x).

(8)

(Total for question = 10 marks)

RocketRevise

(Q11 WMA11/01, Oct 2019)

Q3.

The curve C has equation $y = f(x)$, $x > 0$

Given that

- $f'(x) = \frac{2}{\sqrt{x}} + \frac{A}{x^2} + 3$, where A is a constant
- $f''(x) = 0$ when $x = 4$

(a) find the value of A.

(4)

Given also that

- $f(x) = 8\sqrt{3}$, when $x = 12$

www.RocketRevise.com

(b) find f(x), giving each term in simplest form.

(5)

(Total for question = 9 marks)

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

(**Q07 WMA11/01, June 2022**)

Q4.

Given that $y = 4x^3 - \frac{5}{x^2}$, $x \neq 0$, find in their simplest form

(a) $\frac{dy}{dx}$,

(3)

(b) $\int y \, dx$

(3)

(Total for question = 6 marks)

(**Q01 WMA11/01, Specimen papers**)

Q5.

A curve with equation $y = f(x)$ passes through the point (4, 25)

Given that

$$f'(x) = \frac{3}{8}x^2 - 10x^{-\frac{1}{2}} + 1, \qquad x > 0$$

find $f(x)$, simplifying each term.

(Total for question = 5 marks)

(**Q07 WMA11/01, Specimen papers**)

www.RocketRevise.com

Q6.

Find, in simplest form,

$$\int \left(\frac{8x^3}{3} - \frac{1}{2\sqrt{x}} - 5 \right) dx$$

(4)

(Total for question = 4 marks)

RocketRevise

(Q01 WMA11/01, Jan 2020)

Q7.

Find

$$\int \left(10x^5 + 6x^3 - \frac{3}{x^2} \right) dx$$

giving your answer in simplest form.

(Total for question = 4 marks)

(Q01 WMA11/01, June 2022)

www.RocketRevise.com

Q8.

A curve has equation $y = f(x)$, $x > 0$

Given that

compiled by Ahmed Gul www.RocketRevise.com

- $f'(x) = ax - 12x^{\frac{1}{3}}$, where a is a constant
- $f''(x) = 0$ when $x = 27$
- the curve passes through the point $(1, -8)$

(a) find the value of a.

(3)

(b) Hence find f(x).

(4)

RocketRevise

(Total for question = 7 marks)

(**Q10 WMA11/01, Oct 2021**)

Q9.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(a) Find the equation of the tangent to the curve with equation

$$y = \frac{1}{4}x^3 - 8x^{-\frac{1}{2}}$$

at the point $P(4, 12)$

Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

(5)

www.RocketRevise.com

The curve with equation $y = f(x)$ also passes through the point $P(4, 12)$

Given that

$$f'(x) = \frac{1}{4}x^3 - 8x^{-\frac{1}{2}}$$

(b) find f(x) giving the coefficients in simplest form.

(5)

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

(Total for question = 10 marks)

(Q08 WMA11/01, June 2023)

Q10.

The curve C has equation $y = f(x)$ where $x > 0$

Given that

RocketRevise

- $f'(x) = \dfrac{4x^2 + 10 - 7x^{\frac{1}{2}}}{4x^{\frac{1}{2}}}$
- the point $P(4, -1)$ lies on C

(a) (i) find the value of the gradient of C at P

(ii) Hence find the equation of the normal to C at P, giving your answer in the form $ax + by + c = 0$ where a, b and c are integers to be found.

(4)

(b) Find $f(x)$.

(6)

(Total for question = 10 marks)

(Q07 WMA11/01, Oct 2023)

www.RocketRevise.com

Q11.

A curve C has equation $y = f(x)$, $x > 0$

Given that

- $f''(x) = 4x + \dfrac{1}{\sqrt{x}}$

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

- the point P has x coordinate 4 and lies on C
- the tangent to C at P has equation $y = 3x + 4$

(a) find an equation of the normal to C at P

(2)

(b) find f(x), writing your answer in simplest form.

(6)

RocketRevise

(Total for question = 8 marks)

(**Q011 WMA11/01, Jan 2023**)

Q12.

Find

$$\int \frac{4x^5 + 3}{2x^2} \, dx$$

giving your answer in simplest form.

(Total for question = 5 marks)

www.RocketRevise.com

(**Q03 WMA11/01, Jan 2023**)

Q13.

A curve with equation y = f(x) passes through the point (9, 10).

Given that

www.RocketRevise.com

$$f'(x) = 27x^2 - \frac{21x^3 - 5x}{2\sqrt{x}} \qquad x > 0$$

find f(x), fully simplifying each term.

(Total for question = 6 marks)

(Q09 WMA11/01, Oct 2020)

RocketRevise

Q14.

Find

$$\int \frac{4x^2 + 1}{2\sqrt{x}} dx$$

giving the answer in its simplest form.

(5)

(Total for question = 5 marks)

(Q04 WMA11/01, June 2019)

Q15.

www.RocketRevise.com

Find

$$\int \frac{(3\sqrt{x} + 2)(x - 5)}{4\sqrt{x}} dx$$

writing each term in simplest form.

(Total for question = 6 marks)

compiled by Ahmed Gul www.RocketRevise.com

(Q04 WMA11/01, June 2021)

Q16.

The curve C has equation $y = f(x)$ where $x > 0$

Given that

- $f'(x) = \dfrac{(x+3)^2}{x\sqrt{x}}$
- the point $P(4, 20)$ lies on C

RocketRevise

(a) (i) find the value of the gradient at P

(ii) Hence find the equation of the tangent to C at P, giving your answer in the form $ax + by + c = 0$ where a, b and c are integers to be found.

(4)

(b) Find $f(x)$, simplifying your answer.

(7)

(Total for question = 11 marks)

(Q06 WMA11/01, Jan 2022)

www.RocketRevise.com

Q17.

Find

$$\int 12x^3 + \frac{1}{6\sqrt{x}} - \frac{3}{2x^4}\,dx$$

giving each term in simplest form.

(5)

www.RocketRevise.com

compiled by Ahmed Gul www.RocketRevise.com

(Total for question = 5 marks)

(Q01 WMA11/01, Oct 2021)

Q18.

Find

RocketRevise

$$\int\left(\frac{8x^3}{5}-\frac{2}{3x^4}-1\right)\mathrm{d}x$$

giving each term in simplest form.

(Total for question = 4 marks)

(Q01 WMA11/01, Jan 2022)

Q19.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

The curve C has equation $y = \mathrm{f}(x)$, $x > 0$

Given that

www.RocketRevise.com

- $\mathrm{f}'(x) = \frac{12}{\sqrt{x}} + \frac{x}{3} - 4$
- the point $P(9, 8)$ lies on C

(a) find, in simplest form, $\mathrm{f}(x)$

(5)

The line l is the normal to C at P

(b) Find the coordinates of the point at which l crosses the y-axis.

www.RocketRevise.com

(4)

(Total for question = 9 marks)

(Q05 WMA11/01, Oct 2022)

Q20.

RocketRevise

(i) Find

$$\int \frac{(3x+2)^2}{4\sqrt{x}}\,dx \qquad x > 0$$

giving your answer in simplest form.

(5)

(ii) A curve C has equation $y = f(x)$.

Given

- $f'(x) = x^2 + ax + b$ where a and b are constants
- the y intercept of C is –8
- the point $P(3, -2)$ lies on C
- the gradient of C at P is 2

find, in simplest form, $f(x)$.

(6)

(Total for question = 11 marks)

www.RocketRevise.com

(Q09 WMA11/01, Jan 2021)

Q21.

A curve has equation $y = f(x)$, where

compiled by Ahmed Gul www.RocketRevise.com

$$f''(x) = \frac{6}{\sqrt{x^3}} + x \qquad x > 0$$

The point $P(4, -50)$ lies on the curve.

$f'(x) = -4$ at P,

(a) find the equation of the normal at P, writing your answer in the form $y = mx + c$, where m and c are constants,

(3)

(b) find $f(x)$.

(8)

RocketRevise

(Total for question = 11 marks)

(Q11 WMA11/01, Jan 2020)

Q22.

The curve C with equation $y = f(x)$, $x > 0$, passes through the point $P(4, 1)$.

Given that $f'(x) = 4\sqrt{x} - 2 - \frac{8}{3x^2}$

(a) find the equation of the normal to C at P. Write your answer in the form $ax + by + c = 0$, where a, b and c are integers to be found.

(4)

(b) Find $f(x)$.

www.RocketRevise.com

(5)

(Total for question = 9 marks)

(Q08 WMA11/01, June 2019)

www.RocketRevise.com

Q23.

The curve C has equation $y = f(x)$, $x > 0$

Given that

- C passes through the point P (8, 2)
- $f'(x) = \frac{32}{3x^2} + 3 - 2\left(\sqrt[3]{x}\right)$

(a) find the equation of the tangent to C at P. Write your answer in the form $y = mx + c$, where m and c are constants to be found.

RocketRevise

(3)

(b) Find, in simplest form, $f(x)$.

(5)

(Total for question = 8 marks)

(**Q06 WMA11/01, June 2021**)

www.RocketRevise.com